Tejas Mehtre

ee23btech11059@iith.ac.in | pictures.tejas@gmail.com | +91 8121269805 | LinkedIn | GitHub

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	7.96
Primus Public School	Class XII (State Board)	2023	91.10%
Anand Shiksha Kendra	Class X (CBSE)	2021	95.20%

ACHIEVEMENTS

- Selected for an **international internship** at **Hioki E.E. Corporation, Japan** (2026).
- Won Waveform Wars, the **Signal Processing Hackathon** of Elan & nVision 2025. Trained an object detection **model to accurately identify rooftops** and a MATLAB based spectral subtraction method of **denoising audio.**
- **300+ C++ Problems** (LeetCode) and **5 Stars** on HackerRank **C Section.**
- **Rewarded by the Department of Science and Technology**, Govt. of India through **inspire awards.**

FINAL YEAR PROJECT

FPGA Verification and Framework for NPU Primitives.

Guide: Prof. Amit Acharyya

- **Implemented a DMA-driven AXI4-Stream verification framework** on Zynq UltraScale+ enabling rapid **FPGA testing** of **NPU primitives** (GeLU, SiLU, Systolic array) on a **ZCU104** with end-to-end data validation.
- Implemented a **FSM controlled custom AXI wrapper over the DUT** to receive, process and transmit data.
- Debugged AXI protocol interactions using ILA. Explored **AI Chip Architectures.**
- Wrote **Bare-metal C applications** in Vitis to drive DMA transfers, **manage cache coherency** and **verify output data.**

SKILLS

Programming Languages: C++, Verilog, Python, C, ARMv4 ISA, Shell Scripting

Protocols: AXI, UART, I2C, SPI

Boards: AMD ZCU104 , Raspberry Pi RP2350

Libraries: TensorFlow, SciKitLearn, PyTorch, Numpy, Pygame, Selenium, Matplotlib

Tools: Git, Vivado, ILA, Vitis, LTSpice, MATLAB, Cadence, Altium Designer

PROJECTS

4-point FFT, FIR & IIR filter in Verilog: | *Verilog, Fixed-Point Arithmetic, MATLAB*

- Implemented FFT, FIR, and IIR filter in Verilog **using fixed-point arithmetic for hardware efficient design.**
- Utilized **quantized data to reduce cost of compute** and **verified against MATLAB** output.
- **Pipelined** the FIR filter **to better the throughput.**

RTL for Transformer Accelerator: | *Verilog, Vivado*

- **Developed and verified RTL** for the **bilinear interpolator** and **decoder** required for the same in Verilog for a Reconfigurable and Efficient Transformer Accelerator **made for Autonomous Driving** (RETA-AD).
- Designed multiple **address generation logic's** for the decoder **optimizing FPGA resource utilization.**
- **Reduced fanout** through register duplication for the interpolator **to optimize timing.**

Selective Active Noise Canceller (ANC) for headphones: | *DSP, MATLAB, Algorithms*

- Achieved an **SNR of 23.8 dB** using an adaptive recursive least squares (RLS) filter.
- **Developed from scratch in MATLAB**, the **design features both full suppression and partial suppression mode** which used a programmable notch filter for **multiple frequency isolation.**
- This enabled users to selectively **allow specific ambient sounds such as alarms/notifcations to pass through,** improving situational awareness without sacrificing acoustic isolation.

Design and implementation of a 5-bit SAR ADC | *Hardware, System Design*

- Central circuit included **D flip-flop based SAR logic**, peripherals involved using a **R-2R ladder DAC, comparator.**
- Integrated with **Arduino UNO for clock, power, and set/clear operations.**
- Hands on hardware and **lab equipment experience.**

OTHER NOTABLE WORK

- Rooftop Segmentation using YOLOv8 - TinyML on RP2350 - Pong in Python - Seq2Seq NMT

RELEVANT COURSES

- Digital Systems - Computer Architecture - Data Structures and Algorithms - Operating Systems - Embedded Programming - Machine Learning - Object Oriented Programming - Digital Signal Processing - CMOS VLSI Design - RTL Design & Verification - Semiconductor Device Fundamentals - Analog Electronics - Memory Circuits and System Design - PCB Design for Electronic Systems - Analog IC Design - Probability & Random Processes - Matrix Theory - Vector Calculus

CERTIFICATIONS

- **Kaggle:** Computer Vision, Intro to Deep Learning, Natural Language Processing
- **HackerRank:** C Section
- **MATLAB:** Onramp

EXTRACURRICULAR

- Part of the **Athletic Division** of **National Sports Organisation (NSO)** at IIT Hyderabad.
- **Coordinator** for **Elan & nVision 2025** handling Multimedia and Events (**Position of Responsibility**).
- **Hostel Representative** - Sarabhai Block (**Position of Responsibility**)